

Durability of GFRP bars: A critical review of the literature

Gilbert Nkurunziza, Ahmed Debaiky, Patrice Cousin and Brahim Benmokrane
University of Sherbrooke, Canada

Summary

The research undertaken during the last two decades has shown that one of the potential solutions to the steel-corrosion-related problems in concrete is the use of fiber-reinforced composite (FRP) reinforcement as a replacement for traditional steel bars. Glass FRP (GFRP) reinforcement is gaining more popularity in construction of bridges and in other concrete structures because of its low cost compared to Carbon FRP reinforcement. The durability of these materials, especially under severe environmental conditions, is now recognized as the most critical topic of research. The lack of data on durability of the GFRP reinforcements is a major obstacle to their acceptance on a broader scale in civil engineering.

This paper summarizes the most significant research work published on the durability of FRP bars in the past two decades. A comprehensive review of the literature on the durability of FRP bars indicates a significant increase in the number of studies in this area in the last decade. The durability tests conducted by the authors and others on the latest generation of GFRP bars subjected to stresses higher than the design limits, combined with aggressive mediums at elevated temperatures, have concluded that the strength reduction factors adopted by current codes and guidelines are conservative. These factors were based on limited test results that were carried out on the early generations of the GFRP products, which have now substantially changed.

Key words: durability; fiber-reinforced polymers; composites; reinforced concrete

Prog. Struct. Engng Mater. 2005; **7**:194–209

Published online 10 October 2005 in Wiley InterScience (www.interscience.wiley.com). DOI: 10.1002/pse.205

1 Introduction

Many reinforced concrete structures are exposed to aggressive environments causing durability problems and are, therefore, unlikely to reach their anticipated service life. Corrosion due to chlorides and/or carbonation is the principal cause of deterioration of reinforced concrete structures. The cost of repairs and restoration in the USA, Canada and in the majority of the European countries constitute a high percentage of the expenditure of these countries on infrastructure[1]. In Canada, an estimate shows that more than 40% of all the bridges 40-years old or more and multistorey parking garages are structurally deficient mainly due to corrosion caused by de-icing salts and severe climate[1,2]. In the US, more than 40% of the 574 729 national bridges inspected are classified as structurally or functionally defective because of the corrosion of steel and the damage from freeze-thaw cycling[3]. In addition to these factors, there has been a large increase in traffic loads during the last thirty years. Consequently, the bridges built in the 1950s and

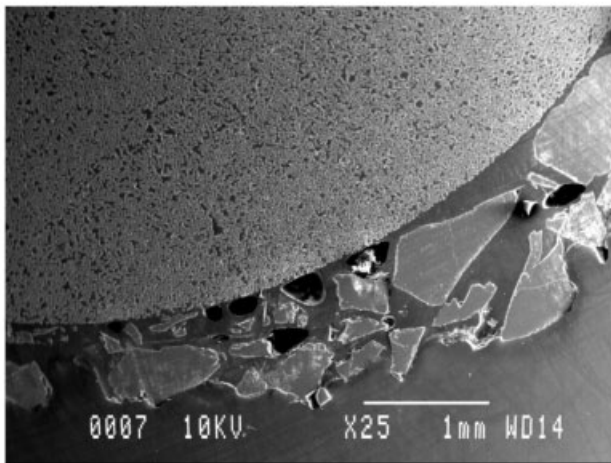
1960s are now subjected to loads higher than their design limits, and, in many cases, twice as much.

The corrosion of steel in concrete structures requires high costs of repair and maintenance. In Canada, the cost of repairs of multistorey parking garages is estimated at more than 6 billion CDN\$[4], while it ranges between 50 and 100 billion US\$ per year in the US[3]. In Europe, the annual cost of repairs is estimated at 1.0 billion £ Sterling. The extent of the phenomena and the economic considerations initiated the development of new technologies in order to reduce corrosion of reinforcement. Various techniques such as cathodic protection, epoxy-coated bars, and galvanized steel were employed. The epoxy-coated steel bars now represent 15% of the North-American steel market. Despite all these efforts, the new protection techniques have not completely succeeded in stopping corrosion. This failure directed the research towards the development of new materials for reinforcement, namely fiber-reinforced polymers (FRP), known under the generic term of composite materials.

The new materials attracted research interest worldwide because of their superior properties, such as light-weight, corrosion resistance, and high mechanical properties. Many international conferences have made it possible to share acquired knowledge, to describe the progress of the work in various centers and institutes of research, and to produce several handbooks and Codes of design. GFRP bars have been used as the primary reinforcement in several bridges around the world. They consist of resin and thousands of fibers, the diameter of each being approximately $15\mu\text{m}$ (Fig. 1). GFRP bars are generally produced by a pultrusion process (Fig. 2). The glass fibers stretched from roving creels are impregnated with liquid resin and preformed through a die. The material is then cured and pulled before being cut to the desired length. In Canada, GFRP bars have already been used as the primary reinforcement in several highway bridges^[5]. The first highway bridge to be entirely reinforced with GFRP bars was recently opened to traffic near the town of Val-Alain, Quebec. The bridge consists of a 200 mm thick concrete deck slab supported on four steel girders. GFRP was used as longitudinal and transverse reinforcement in the slab and barrier walls. Fig. 3 shows the construction of the bridge.



(a)



(b)

Fig. 1 Sand-coated GFRP bars: (a) bars; (b) cross section of the material observed by SEM showing the sand coating and the glass fibers

1.1 SIGNIFICANT ISSUES

The research undertaken during the last two decades has shown that one of the potential solutions to the problems related to steel corrosion is the use of FRP reinforcement as a replacement for traditional steel reinforcing bars. Glass FRP (GFRP) reinforcement is gaining in popularity for construction of bridges and for other concrete structures because of its low cost compared to carbon FRP (CFRP) reinforcement^[6–11]. The durability of these materials, especially under severe environmental conditions, is now recognized as the most critical topic of research. The lack of data on durability of the GFRP reinforcements is a major obstacle to their acceptance on a broader scale in civil engineering^[12].

The extensive research performed on FRP reinforcements led to the publication of codes and handbooks on the use of the FRP products in the reinforcement and the rehabilitation of new and existing bridges and structures. Design codes in North-America and Japan^[2,13–16] provide strength reduction factors or material factors to account for the effects of environmental conditions, creep from

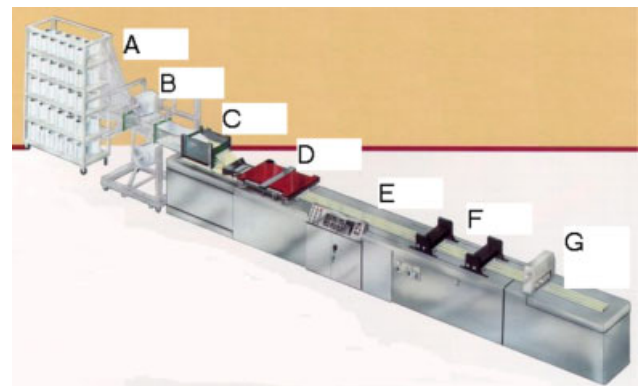


Fig. 2 Pultrusion process showing the different steps of bar manufacturing: (A) Roving; (B) Stretching; (C) Resin bath; (D) Preform and heating die; (E) Bars; (F) Caterpillar pull; (G) Cutter



Fig. 3 Construction of the GFRP-reinforced concrete deck slab in Val-Alain Bridge on Highway 20 East, Quebec, Canada, October 2004.

sustained loads, and fatigue from repeated loads. The Canadian Codes seem to be less conservative with respect to the allowable stresses, but restrict the use of certain types of FRP to certain conditions [17]. The development of a coefficient of environmental reduction (E_c) was a simple solution to overcome the difficulties caused by the absence of data on the long-term durability of the FRP. Several codes require an environmental reduction coefficient of approximately 0.7. When the environmental effects are combined with sustained load, the total coefficient of reduction recommended by the ACI 440.1R-03 guide ranges between 0.14 and 0.16[13]. This major factor makes it impossible to fully utilize the superior properties of the new generation of the FRP products that may not be as vulnerable as their predecessors to environmental effects.

Indeed, the reduction coefficients are considered to be very conservative and are being criticized by many experts and engineers, because the coefficients are based on the behavior of the early generations of FRP products that were of lower quality, and on limited data on their long-term behavior.

2 Review of literature on durability of GFRP reinforcements

This review comprises two principal parts. In the first part, general considerations on the mechanisms of degradation of FRP bars and the details of investigations on durability are presented. After this review on the characteristics of durability of FRP bars, specific information concerning the environmental degradation of the components of FRP bars (fibers and resin) and their interface are given. In the second part, methodologies for accelerated testing and those proposed for the prediction of the lifespan of FRP bars will be discussed.

One of the major challenges for researchers that also represents an obstacle to an increased use of these materials in the civil engineering infrastructures relates to durability. During the last two decades, several studies were carried out on the parameters affecting durability and the long-term behavior of FRP materials used in civil engineering applications. However, the data available on the durability of the FRP bars are dispersed, sometimes contradictory and often improperly documented[18–20].

Accelerated tests were carried out at the University of Iowa by Porter and Barnes[21] to determine the long-term tensile strength of GFRP bars. The FRP bar samples were exposed to an alkaline solution at a high temperature (60°C) for periods of 2–3 months, which represents 50 years of exposure in a Northern climate. The tensile tests carried out on three types of GFRP bars resulted in residual strengths of 34, 52, and 71% compared to the original tensile strength.

In 2002, tests carried out by Sen *et al.*[22] showed that GFRP bars exposed to an alkaline solution of pH 13.5 could lose 70% of their tensile strength after 9 months under a constant loading of 10% of their initial strength. Nkurunziza *et al.*[23] conducted tests on GFRP bars (9.5 mm diameter) under the combined effect of sustained loadings (30 and 40% of the ultimate strength) and immersed in chemical solutions (de-ionized water and an alkaline solution). In addition, as an accelerating ageing condition, the GFRP specimens were subjected to a temperature varying between 55 and 75°C for 60 days. The results obtained from bars conditioned in the de-ionized water and the alkaline solutions showed only a loss of 4 and 11% of their original strength, respectively.

The tensile tests carried out by Gaona[24] on GFRP bars showed that tensile strength decreases with time when the bars are in direct contact with solutions simulating the interstitial solution of the concrete. Losses up to 24% were measured for bars conditioned for 50 weeks in an alkaline solution with high pH value (12) at a temperature of 350°C. The elastic modulus of the bars tended to increase with time, and an average increase of 9% for the 50-week conditioning was recorded.

Until a few years ago, there were no procedures or standard test methods to evaluate the durability of the FRP products as reinforcement for the civil engineering structures. This complicates the comparisons between the results obtained by various researchers. In 2003, Committee ACI 440 K proposed a testing method to evaluate the durability of FRP reinforcing bars under loading and immersed in an alkaline solution[25]. A high temperature up to 60°C could be used to accelerate ageing.

The durability of a material or a structure is defined as its ability to retain its original properties with time under load. A more extended definition related to reinforced concrete is the ability to resist cracking, chemical degradation, delamination, wearing, and similar symptoms of ageing with time, under the conditions of sustained loads and under the specified environmental conditions[26]. Moreover, the durability of a composite material is related not only to the strength of its constitutive materials (fibers and matrix), but also to the integrity of the interface between these two components while ageing. A deterioration of this interface reduces the transfer of the loads between fibers, and thus weakens the composite material.

Significant research on the durability of composite materials in aerospace and aeronautics was carried out in the last thirty years. It was proven that moisture, alkalinity, and temperature are the main parameters affecting the durability of composite materials. The moisture absorbed by the composites, combined with the temperature of exposure, induces stresses in the material that damage the fibers, the matrix, and their interface, thus gradually decreasing

the strength of the composite material with time. The glass transition temperature (T_g) of the matrix is also a key parameter, since it marks a point beyond which significant changes in the properties of the matrix occur and considerable reduction of the mechanical properties takes place.

In civil engineering applications, the study of the durability of composite materials is a relatively recent subject of research. Such studies began in the late 1980s on compound products that were altered to fit for these new applications. A detailed review by Benmokrane *et al.*[27], regarding the environmental effects on the durability of the composites in civil engineering, indicates that the composites can be affected in different ways according to their composition and the conditions of exposure. The effects of moisture as well as alkaline and acid solutions were recognized as the most critical parameters[3,28–31]. The authors also pointed to the need for developing accelerated test methods that can simulate long-term behavior of these materials within a practical time frame in the laboratory.

An interesting summary of the environmental durability of the FRP material in the field of civil engineering was furnished by Karbhari[32], where the various phenomena of degradation of the material were presented. The authors mainly linked the degradation to matrix-related problems such as deterioration of the fibers/matrix interface. The authors also proposed coefficients of reduction that express the drop in strength of the composite materials after short and long terms of exposure. For a long-term exposure, these coefficients varied between 0.25 and 0.5 for the GFRP composites, and were 0.9 for CFRP composites.

The analytical and experimental studies of the long-term behavior of composite materials can be classified into two categories[33]. The first one evaluates the residual properties of the different composites after exposure to various environments without loading. The second category includes the effect of the sustained loading. Several experimental and analytical studies were reported in the literature for the residual performance of the samples exposed to various environmental conditions without loading[3,27–30,34–38]. These tests were carried out under the service conditions, or using accelerated agents, such as elevated temperatures to expedite the degradation or ageing. The results of these studies showed that the materials tested were likely to lose their properties to various degrees according to the type of environment used. Their resistance to the chemical attack depended highly on the matrix type.

Chu and Karbhari[39] completed a study on the characterization and modeling of the effects of moisture and alkalis on GFRP bars (E-glass/polyester). The results of this study showed that several factors contribute to the degradation of the composite material exposed to the various

environments. The prevailing mode of failure can be linked to the rupture of fibers, cracking of matrix, loss of bond at the interface, or to any combination of the three. The moisture content of the surroundings influences the micro-cracking of the resin and accelerates the deterioration of fibers. The use of a very hydrophobic resin can slow down the mechanisms contributing to the degradation of the composite. However, the influence of the environment on each component of the composite material must be well-defined in order to predict the long-term behavior of these materials accurately. All these studies showed that moisture can affect the service life of GFRP bars, and that the rate of degradation increases with temperature.

Although these tests are considered to be important in the evaluation of the long-term performance of composite materials, they do not answer the principal question about the combined effect of load and environment. Very few studies on this combined effect were found in the literature. Studies carried out on FRP exposed to an alkaline environment in the presence of loading at various temperatures are limited[33,40–46]. The stress levels used in these tests exceeded the levels expected at service conditions in order to bring the samples to failure relatively quickly. Elevated temperatures were used to accelerate the degradation of the material so that the long-term performance of the materials could be extrapolated from the short-term data.

Another study on the environmental durability of the GFRP under the combined effect of moisture, temperature, and sustained stress for a short duration[26] showed that the durability of the composites is affected by water immersion and the increase in water temperature, and to a lesser extent by an increase in the applied stress.

In 2004, ISIS Canada approved a project to study the performance of the glass fiber-reinforced polymer (GFRP) that has been used in many demonstration structures across Canada[47]. The objective of the study was to provide the engineering community with the results of the performance of GFRP materials that have been exposed to a concrete environment in built structures and to calibrate the CHBDC code performance factors for GFRP reinforcement. Core specimens of GFRP reinforcement were collected from five field demonstration projects across Canada. Laboratory tools, such as Scanning Electron Microscopy and Energy Dispersive X-ray (EDX), Optical Microscopy, Differential Scanning Calorimetry, and Infrared Spectroscopy, were used to determine the degradation of GFRP in concrete structures.

The results of these studies indicated that there was no visible degradation of the GFRP reinforcement (rods and grids) in the concrete environment in real-life engineering structures exposed to natural environmental conditions for durations of 5–8 years. The EDX analyses indicated no alkali ingress in the

GFRP reinforcement from the concrete pore solution. The conclusion of the project was that under tension GFRP reinforcement is durable and highly compatible with the concrete material. Also, the team concluded that the CHBDC is conservative; therefore, the material performance factors of the GFRP should be increased. This change has been incorporated in the new addendum for the CHBDC^[48].

3 Degradation mechanisms of FRP

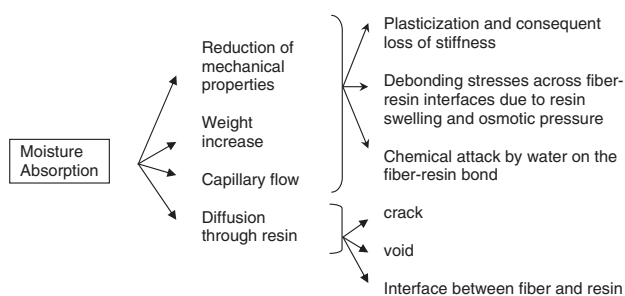
3.1 GENERAL CONSIDERATIONS

When a load is applied to the FRP bar, the fibers carry the load while the resin transfers the stresses between fibers. The resin also protects the fibers against the attack of harmful ions coming from the environment, in particular OH⁻ ions of the interstitial solution of the concrete. Degradation of the FRP is a result of the degradation of one or more of its components, as well as interface between them. This complicates the mechanism of degradation of the FRP reinforcement compared to that of steel.

Since the mechanical properties of the GFRP bars are controlled by the fibers^[49], if fibers do not deteriorate, the GFRP bars will resist the mechanical load. If the resin is attacked and deteriorated, the surface of the fibers starts to break, progressing towards the interior of the bar, leading to a reduction in the bar strength. The interface between glass fibers and the resin controls the resistance of the GFRP bars to the alkalis^[50]. The improvement in the strength of the interface bond effectively increases the resistance of the bar to alkalis. The details of the degradation of the GFRP bar components are discussed in the following sections.

3.2 EFFECT OF MOISTURE ON GFRP REINFORCING BARS

The migration of moisture in FRP composites has been one of the most widely studied subjects during the past several decades. The overall effect of moisture absorption on composites is given in the following pattern:



Depending on the type of resin matrix used and surrounding environment, moisture diffuses into composite materials to varying extents and at various

rates. This diffusion process occurs at the molecular level. Moisture can be absorbed by capillary uptake along any preexisting crack or interface between the fiber and resin matrix. The effect of moisture on composites is a mass uptake, followed by the plasticization of the matrix and a decrease in the glass transition temperature. Moisture can act as a plasticizer, thus reducing the matrix modulus. The literature has also shown that moisture can increase crack density, which would also have an effect on the modulus^[51]. Moisture can attack the fiber during loading causing stress-corrosion cracking. Cracks can form at the fiber–matrix interface, thus decreasing the stress transfer of the composite

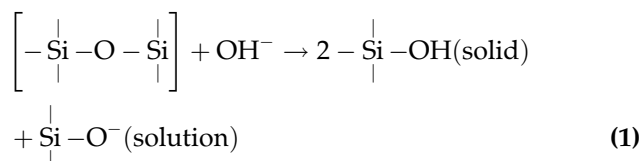
3.3 DEGRADATION OF GLASS FIBERS

The glass fibers are the fiber of choice for civil engineering applications because of their low cost. GFRP reinforcements do not corrode in a chloride environment due to their non-metallic nature; therefore, the following section will discuss only the mechanisms of their degradation in alkaline and in very wet mediums. Indeed, glass fibers can degrade in the presence of heavy moisture, acids, or alkaline solutions. The literature indicates that the most critical degradation takes place in alkaline solutions. Many studies carried out on the resistance of the GFRP bars to alkalis^[52] showed that the alkaline medium of the concrete constitutes one of the most critical environments for glass fibers.

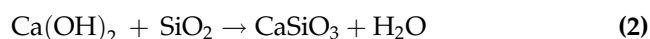
At present, the principal concern for GFRP bars is the potential degradation caused by the interstitial solution of the concrete in which the pH can reach 13.5. Certain concretes can comprise voids due to a bad compaction or the accumulation of bleed water; these voids can represent between 1 and 10% of the volume of the concrete and constitute a privileged flow of the alkaline interstitial solution^[53].

In an alkaline environment, the glass fibers are vulnerable to degradation, following a combination of mechanisms, such as hydrolysis and leaching. Several studies have been published on the mechanisms of degradation of GFRP reinforcements and on the environmental effects^[22,29,43,54]. The degradation of GFRP bars exposed to an alkaline solution is responsible in part for the losses in elastic modulus and tensile strength.

Yilmaz and Glasser^[55] explain that the attack of glass fibers is by the process of dissolution of silica (SiO₂) by alkaline ions (OH⁻), according to eq. (1), where the SiOH byproduct forms as gel on the surface of glass fibers while the SiO byproduct forms as a solution.



This dissolution causes a loss in the weight and diameter of the fiber, and consequently a loss of strength and elastic modulus. Another mechanism of degradation resulting from the reaction of silica with the alkaline medium is expressed by eq. (2)[43]:



Studies have also shown that degradation of the GFRP bars in an NaOH solution is more severe than that in concrete. This severe degradation is due to the higher alkalinity of NaOH and the high mobility of OH⁻ ions[43,56]. Moreover, the resistance of the glass fibers to alkalis also depends on the composition of the fibers. The addition of alumina to the glass fibers increases their resistance to water, but decreases their resistance to alkalis[53]. By adding a certain ratio of zirconium dioxide (ZrO₂) to glass fibers, the resistance to alkalis increases[54,57]. This produces AR (or Alkali Resistant) fibers. When the fibers are zirconium dioxide free, they again become sensitive to the alkalis attack. The majority of the studies showed that the alkaline environment is more critical than water at the beginning of degradation, but after a certain time both effects become almost similar[39].

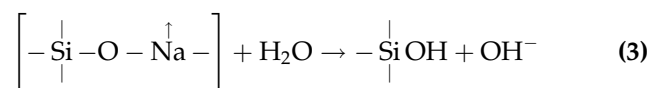
A study into the diffusion of moisture on various types of FRP bars immersed in various solutions[56] showed that in the alkaline solution, the temperature effect is limited by the maximum moisture content, and that the diffusion of the solution into the vinyl ester resin was less when compared to the diffusion into the polyester resin.

However, in a composite material, the fibers are integrated in the polymeric matrix, and any degradation of fibers implies the transport of the corrosive agent through the matrix. Consequently, a composite material with an impervious matrix can have a good resistance to the alkalis attack. The use of high-performance resins, such as the vinyl ester, and coupling agents on the surface of fibers enormously improves the environmental protection of the fibers[50].

Moisture diffuses inside organic polymers to varying degrees, according to the changes in the mechanical, thermo-mechanical and physico-chemical characteristics. The principal effect of absorption appears on the matrix through hydrolysis or plasticization. In the case of glass fibers, degradation is initiated through the extraction of ions by moisture, thus deteriorating the chemical structure of the fibers. Once exposed to alkaline solution, acid, or even aqueous mediums, the glass fibers containing alkalis (Na⁺, K⁺, Ca²⁺, Mg²⁺, Mg³⁺) become prone to chemical attack. Four mechanisms of degradation of fibers due to water are illustrated by the equations of Mercier and Marechal[58].

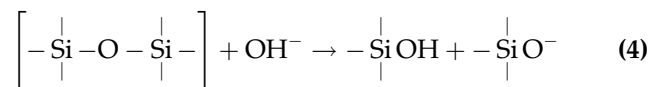
The first and the most important reaction of dissolution of fibers in water is the diffusion of the alkalis ions out of the glass structure (eq. (3)). This process is also called leaching in the literature, which

is exacerbated by the quantity of alkalis present in the glass structure and the temperature of the wet environment.

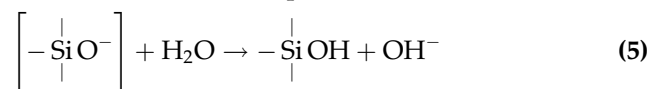


In eq. (3), the connection between sodium–oxygen and the interface is broken, and the remaining oxygen atom collects a hydrogen ion, H⁺. The new configuration causes additional stresses on the surface of the glass involving the formation of the cracks. As a result, the load applied widens the cracks, which propagate causing the failure of the sample. In a similar way to the reaction of Na⁺ ion, ionic exchange leads to the replacement of other ions, such as Na⁺, K⁺, Ca²⁺, Mg²⁺, or Mg³⁺, close to the surface of glass by protons. The glass fibers that contain more of these ions in their chemical constitution are more likely to degrade[57].

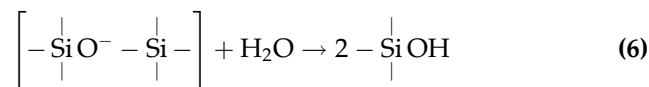
The second byproduct of eq. (3) is silicon hydroxide (SiOH), which is a gel layer formed at the interface of glass and water. This layer is less dense than the original structure of glass and can transport water or the alkaline solution more easily, which accelerates the process of degradation[57]. This mechanism of degradation deserves a detailed discussion because it constitutes the second most important cause in the process of dissolution. If the pH of the solution exceeds 9, the connections Si–O–Si and Si–O–R (R = Na or K) are affected[59]. The process in which the hydroxide ion breaks the Si–O–Si connection is shown in eq. (4):



The byproduct of the preceding equation, SiO⁻, can dissociate another water molecule of the solution mixture as shown in eq. (5).



As can be noted in the three preceding equations, there is a formation of excess hydroxides, which consequently increases the pH value of the solution. These ions attract the surface of the fibers as shown in eq. (6) and generate cracks, which degrade the strength of the bar and can result in a premature fracture or a rupture of the glass fibers[3,22,40,57,60].

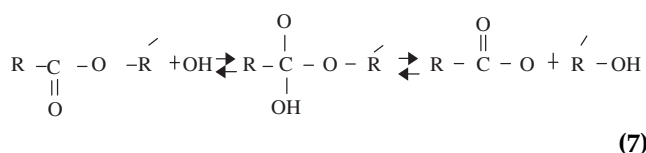


The process of degradation of glass fibers is temperature dependant, because the rate of diffusion depends closely on the temperature.

3.4 DEGRADATION OF THE POLYMERIC MATRIX

The chemical durability of the thermoset resins is controlled mainly by the chemical nature of the

polymer chain. The structure of the polyester and vinyl ester resins consists of long molecular chains formed by the polymerization of unsaturated organic compounds. In the case of polyester resin, the long exposure to OH^- and Cl^- ions involves a breakage of the bond between the oxygen and carbon atoms in the molecular chains. This breakage induces micro-cracks and fractures in the matrix structure and involves a loss of strength with a fast increase in the moisture diffusion. The reaction of the polymer chain with a strong alkaline environment is shown in eq. (7)[55]:



The reaction with the vinyl ester matrix is formed when the double connection between the carbon atoms in the molecular chain is transformed into a simple bond, thus forming a point of reticulation. When this resin is exposed to OH^- and Cl^- ions, an initial chemical reaction takes place outside the simple bond. Consequently, the bond between the carbon atoms, which forms the molecular chain, does not break and preserves the strength and the durability of the matrix. Therefore, vinyl ester epoxies are very resistant to chemical attack, which improves the environmental resistance of the bars made with these resins[61]. This chemical performance of vinyl ester in an alkaline environment was confirmed by several studies[3,54,61,62].

The mechanism of failure prevailing in FRP reinforcements exposed to an alkaline environment and subjected to sustained loading is the delamination at the glass fiber/resin interface as shown in Fig. 4. Resin cracking, creating a pathway for moisture and ionic species, may also be observed (Fig. 5)[63].

3.5 DEGRADATION OF THE FIBER-MATRIX INTERFACE

The fiber-matrix interface is a heterogeneous area between the matrix and the fibers and is a few

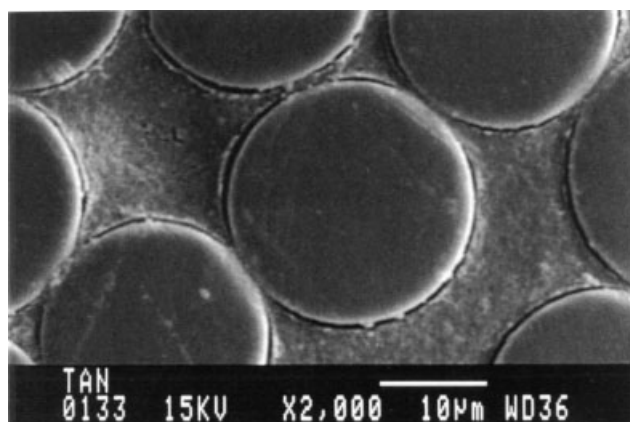


Fig. 4 Delamination at the fiber interface stamps (after[27])

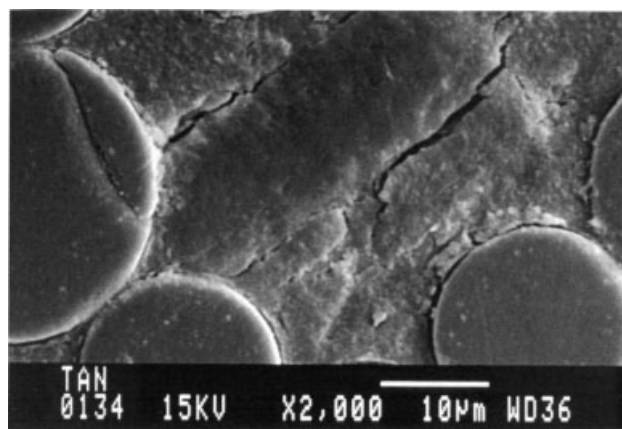


Fig. 5 Cracking of matrix (after[27])

micrometers thick. This interface plays a critical role in the transfer of load between fibers and matrix[64]. The use of a coupling agent on the fiber surface improves the strength of the interface, and protects the surface of the fibers against environmental attack or reaction with moisture or alkalis. However, the chemical bond between the coupling agent and the surface of glass fibers is not stable in the presence of moisture and alkalis[63,64]. When moisture and the alkalis diffuse in the composite, this bond is destroyed gradually, causing damage to the interface. Very few studies have been carried out on the thermo-mechanical properties of the interface because of the difficulty of the experimental procedures. Thomason[64] carried out a series of analytical studies to examine the composition of the layers on the glass fiber surface. The results obtained indicated that the quantity of coupling agent on type-E glass fibers varies between 0.2 and 1% of the weight of fibers and that a good part of this percentage can be extracted by hot acetone. The strength of the interface of the composite was directly related to the quantity of coupling agent that remains on the surface of the fibers.

Combined with chemical attack, a high level of sustained mechanical loading increases the degradation of the fibers/matrix interface. Different modes of damage can occur at the interface, such as (1) osmotic cracking of the matrix, (2) differential swelling of the interface, and (3) separation.

3.6 THERMAL EFFECT

Due to thermal softening, the elastic modulus and strength of an FRP bar decrease with the increase in temperature when tested at high temperature under sustained load[65]. In a polymeric composite, the matrix properties are more affected by the increase in the temperature than the fiber properties. The axial mechanical properties (strength and modulus) of a linear composite with fibers oriented at 0° to the load remain unchanged with the increase in the temperature, while the transverse properties are

reduced when the temperature approaches the glass transition temperature (T_g).

In the short term, an elevation in temperature of between 30 and 40°C does not significantly affect the strength or the elastic modulus of the majority of commercial fibers[65]. However, prolonged thermal ageing at a high temperature combined with sustained loading can cause deterioration in the properties of the matrix[50,66].

3.7 CREEP AND RELAXATION OF GFRP BARS

Phillips[67] indicated that the process of stress rupture is controlled by two conditions: stress and environment. The effects of both mechanical load and the environment must be added to estimate the long-term performance of the material. Tests must be carried out at a sufficient stress level to be able to determine the role of the environment in the time to failure.

Pioneer research carried out on the creep and relaxation of the FRP bars[19,57] showed that the creep deformation of GFRP bars was twice that of steel. However, the low modulus of elasticity of GFRP bars can be an advantage, particularly in prestressing applications, since it produces less reduction in the prestressing force due to shrinkage, creep, and other structural deformations.

In 2002, Greenwood[68] performed tests on GFRP bars to predict their long-term performance. The tests were carried out on 6.4 mm diameter pultruded bars made of high strength or standard E-glass fibers in a polyester matrix. The bars were submitted to a combination of sustained loading and one of three types of surrounding environments: air, salt water, and concrete pore water solution. The author claimed that the conditions of exposure were equivalent to 50 years of service. The results of this study showed that in the air, the two types of fibers have similar performance with a residual ultimate strength equal to 45% of their initial strength. In salt water, the residual strengths were 37 and 27% of the initial strength of the high strength and the standard E-glass fibers, respectively. The alkaline environment was the most aggressive among the three, with residual strengths equal to 25 and 15% of the initial strength of high strength fibers and standard E-glass fibers, respectively. However, it is important to note that these results were obtained with polyester resin, which is much less durable than vinylester resin.

The failure process of samples subjected to sustained stress and environmental conditioning is very complex (Fig. 6). High levels of stress can form micro-cracks in the matrix of the FRP bars. These micro-cracks create a network for the penetration of corrosive agents from the surrounding medium towards the core of the bar. When the level of stress is sufficiently low, the visco-elastic behavior of the matrix prevents the formation of cracks. This means

that there is a stress threshold below which the micro-cracking in the matrix cannot occur, and capillary action dominates the degradation of the bars[69].

At the University of Sherbrooke (Quebec, Canada), two Masters and one Doctoral theses concerning with the durability of GFRP bars have been recently completed[70–72]. These studies were carried out on first-generation GFRP bars. In addition, two other Doctoral theses focusing on the durability of E-glass/vinyl ester FRP bars in alkaline and de-ionized water under sustained tensile stress (or no stress) were also completed[73,74]. Bars of different diameters (9.5, 12.7, and 15.9 mm) were tested under ambient and elevated temperatures up to 60°C for periods of up to 14 months. It was concluded that the creep strain in the 9.5 mm bars was less than 5% of the initial value after 10 000 h of sustained tensile loading. This value was obtained under high tensile stress of 38% of the guaranteed tensile strength. Fig. 7 shows the change in strain in bars with time.

Wang[74] conducted 330 accelerated aging tests on 9.5, 12.7, and 15.9 mm diameter E-glass/vinyl ester reinforcing bars. The samples were conditioned in simulated pore water solution (pH 12.6–12.8) and distilled water at 23, 40 and 60°C for 150 and 300 days. At the end of 150 and 300 days, residual tensile and

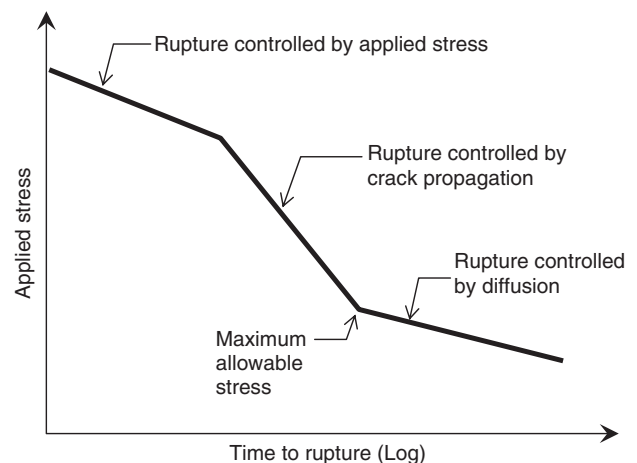


Fig. 6 Behavior to the rupture under environmental stress of reinforced glass fiber composites (after[69])

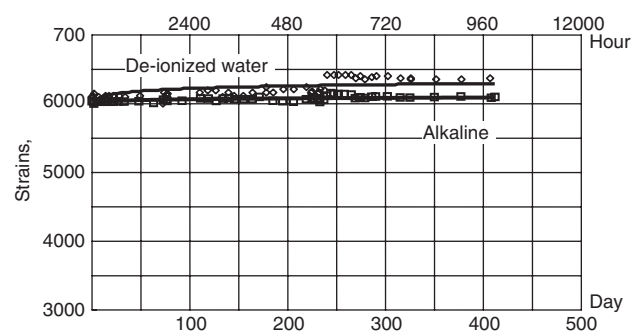


Fig. 7 Creep effect on 9.5 mm GFRP bar in alkaline and de-ionized water solutions for 10 000 h under sustained stress of 38% of the guaranteed tensile strength (after[23])

flexural properties (strength and stiffness) were evaluated. In addition, various microstructural analyses were conducted to identify the effect of moisture, alkaline, and temperature on the degradation mechanisms of E-glass/vinyl ester reinforcing bars. It must be noted that the RH in real concrete environment is lower than 100%; therefore, in an outdoor concrete environment, degradation is likely to take place at a lower rate than that of the samples immersed in the simulated concrete pore water solution.

Obvious changes in the hydroxyl group peaks were observed in samples aged in distilled water and alkaline solution at 60°C. These samples were taken from the parts, that were close to the surface coatings. For all 150-days-aged samples, no defects or holes were observed on the fiber of the samples. Holes or defects were observed on surface coatings, these holes increase with increased temperature. For all 300-days-aged samples, fiber degradation and matrix cracks were observed when samples were aged in alkaline solution and distilled water at 60°C.

Another objective of this research was to predict the service life of selected E-glass/vinyl ester reinforcing bars. The service life prediction was focused on accelerated aging test results of 9.5 mm bars, which were conditioned in distilled water at 23, 40, 60, and 80°C for 30, 100 and 120 days. Attempts were made to use the predictive methodology given in the FHWA material specification and 'Time Shift Factor' methods.

It was concluded that for these specimens the residual flexural strength data showed a dependence on both duration and temperature of conditioning. Changes in the flexural modulus of elasticity were not significant, because unlike strength the modulus is not sensitive to localized deterioration. The degradation in strength of the rebar after environmental exposure is mainly due to a change in the stress transfer efficiency between the fibers and matrix within the composite.

Sirois[70] studied the durability of GFRP bars under tension in an alkaline environment. It was concluded that polyester-based bars tested for creep behavior are very sensitive to alkalis when a stress limit ranging from 25 to 40% of the ultimate strength is exceeded. The author suggested that, for good durability of the GFRP bars, the loads applied must remain lower than 25% of their ultimate tensile strength. It should be noted that the bars used in this study were provided by Pultrall Inc. (Isorod)[75], International Grating (Kodiak), Marshall Industries Composite (C-Bar), and Hughes Brothers. These were first-generation products that have been improved significantly since the time of this research.

Another study on the evaluation of the durability of GFRP bars in an alkaline environment was completed in 1999[71]. The tests were carried out on GFRP bars provided by the same companies as for Sirois[70]. The

diameters of the bars varied from 9.5 to 12.7 mm. It was concluded that under a sustained tensile stress of 30% of the ultimate strength and in an alkaline solution simulating the concrete pore water, the 'corrosion' mechanism of GFRP reinforcements is mainly related to the damages occurring at the fiber/matrix interface. However, in concrete and under a high level of stress, i.e. 55% or more of the ultimate strength, the matrix and fiber fracture under stress was the dominating degradation mechanism.

Laoubi[72] evaluated the durability of GFRP bar in reinforced concrete beams subjected to the combined effect of sustained loads and freeze/thaw cycling. The applied sustained loads, in the form of four-point bending, corresponded to approximately 1.4 times the cracking load, whereas the freeze/thaw cycles varied between -20°C and +20°C with 50% relative humidity. After the conditioning, the beams were tested to failure in four point bending. The following conclusions were drawn from this study:

- The freeze/thaw cycles did not have a significant effect on the flexure behavior of GFRP-bar-reinforced concrete beams after 100, 200, or 300 cycles;
- The combination of constant loads and freeze/thaw cycles did not have any significant effect on the flexure behavior of the beams tested;
- A sustained load corresponding to 1.4 times the cracking moment did not show any significant effect on the flexure behavior of the beams after 7, 13, or 26 weeks.

4 Accelerated tests for long-term performance evaluation of FRP

There are several prediction models for the residual strength of GFRP bars currently available in the literature[3,18,31,33,41,42].

The methodology used for the accelerated ageing tests is based on the Arrhenius principle, which uses high temperature as the accelerating agent for the degradation of the material in a given environment[33].

The Arrhenius empirical model is one of the oldest and most widely used models to predict the time of failure of a material as a function of temperature. This model is given as

$$t_f = A \exp\left(\frac{\Delta H}{kT}\right) \quad (8)$$

where t_f is the real time of failure, T the temperature in Kelvin at the time of failure, k the Boltzmann constant ($8.617 \times 10^{-5}/K$), A the acceleration factor determined graphically from the acceleration tests, and ΔH is the activation energy factor.

The value of ΔH , which varies from 0.3 to 1.5, depends on the failure mechanism and the type of the material used. The acceleration factor between two

temperatures increases exponentially with the increase of ΔH .

The acceleration factor between a high temperature T_2 and a lower temperature T_1 is based on the following equation:

$$A = \exp\left(\frac{\Delta H}{k} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right) \quad (9)$$

By using the value of k given above (Boltzmann constant), eq. (9) can be re-written as a function of the temperature T (in Celsius):

$$A = \exp\left(11605\Delta H \left(\frac{1}{(T_1 + 273.16)} - \frac{1}{(T_2 + 273.16)}\right)\right) \quad (10)$$

In eq. (10), ΔH is the only unknown parameter. The acceleration factors have been calculated for different values of energy of activation ΔH with $T_1 = 23^\circ\text{C}$ and different values of T_2 higher than T_1 by Nkurunziza *et al.*[46].

Litherland *et al.*[33] correlated the data of accelerated ageing tests in the laboratory with the natural ageing of glass fibers in an alkaline environment. The following factors were taken into account in Litherland's method: (1) the annual average temperature was used; (2) no load was applied to the sample; (3) the FRP manufacturing process and the type of resin slow down the fiber degradation. Consequently, a shift factor must be introduced on the time scale when the data are interpreted.

Using Arrhenius' principle, Vijay and Gangarao[31] proposed a correlation formula between the accelerated ageing in the laboratory and the natural ageing on the field. The formula developed for GFRP reinforcements conditioned in aqueous and alkaline solutions is as follows:

$$N/C = 0.098 * e^{0.0558 * T} \quad (11)$$

where N is the natural age in days, T the temperature of conditioning in the laboratory ($^\circ\text{F}$), and C the number of days of exposure in the laboratory at temperature T . This correlation is presented in Fig. 7.

Eq. (11) was developed for the climatic conditions of Northeastern US, specifically in the state of West Virginia (average annual temperature of 11.7°C). Fig. 8 shows the equivalence of the lifespan of the GFRP bars in the field as a function of the accelerated ageing in the laboratory according to eq. (11). According to this equation, 104 days of exposure at 60°C correspond to 69 years of service life in standard concrete subjected to an annual average temperature of 11.7°C .

Another model relating accelerated ageing data to natural ageing was presented by Porter and Barnes[21]. This relationship links the different annual average temperatures by an accelerating factor, which leads to eq. (12) (coefficient of correlation $r^2 = 0.993$). Eq. (12)

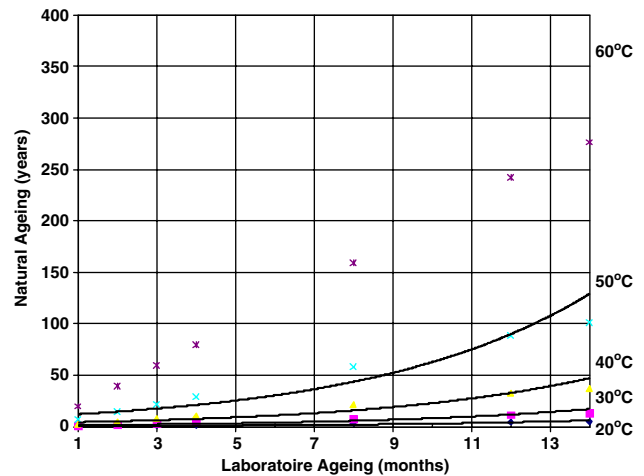


Fig. 8 Correlation between the accelerated ageing and natural ageing (after[31])

was obtained using the annual average temperature of Des Moines (Iowa, USA)

$$\text{AGE} = 1.025 * e^{0.0935(T)} \quad (12)$$

where T is the temperature in Fahrenheit.

The expression presented was adapted to the climatic conditions of Montreal, Canada, and seems to be more optimistic. It results in the assumption that 104 days of laboratory exposure at 63°C simulates 160 years of life in an outdoor condition with an annual average temperature of 6.7°C .

5 Concepts of design of FRP-reinforced concrete

FRP bars have high tensile strength and generally low elastic modulus compared to steel reinforcement. Consequently, the design of concrete elements reinforced with FRP is controlled by the serviceability criteria such as deflection and crack width. However, the basic assumptions used to design steel-reinforced concrete structures still apply to FRP-reinforced concrete structures. These include the linear strain distribution over the depth of the section and the perfect bond between bars and concrete.

The inherent differences in mechanical and mineralogical properties between steel and FRP reinforcement necessitate the development of suitable standards for FRP. Several countries such as Japan, Canada, the US, and the European Union have already established design and building guidelines for the use of FRP bars as internal reinforcement for concrete structures. Based on experimental results, these documents provide general information on the history and the use of FRP reinforcements, the description of their properties, and design and building recommendations for FRP-reinforced concrete structures. The current North-American codes and guidelines are ISIS-M03-01, ACI 440.1R-03, CAN/SCA-S6-00, and CAN/SCA-S806-02[2,13–15].

These documents ensure an acceptable level of safety by introducing reduction factors to account for the effects of the variation of material properties, sustained load, creep, and environmental effects. These factors must be associated with the original material properties to give a total level of safety that defines stress limits to ensure an acceptable long-term performance of the bars. These factors are given in Table 1 for the GFRP bars according to ACI 440.1R-[13], CAN/SCA-S6-00[14], and JSCE[16] documents. Table 2 shows the most common properties of GFRP bars in comparison with traditional steel bars as given by ACI 440.1R-03[13].

5.1 REDUCTION FACTOR FOR ENVIRONMENTAL CONDITIONS

The properties of materials, such as the guaranteed tensile strength, must be regarded as initial properties that do not include the long-term effects from environmental exposure. Consequently, the properties of the composite materials must be reduced by taking into account the types and levels of environmental exposure. According to the ACI 440.1R-03 guide[13], the design tensile strength is given by the following expression:

$$f_{fu} = C_E f_{fu}^* \quad (13)$$

where f_{fu} is the design tensile strength of the FRP in MPa, C_E the environmental reduction factor that depends on the types of fibers and environmental

exposure (the values of this coefficient are provided in Table 3), and f_{fu}^* the guaranteed tensile strength of the FRP reinforcement, defined as the average tensile strength (f_u) less than three times its standard deviation (σ) in MPa.

The maximum allowable deformation is calculated as follows:

$$\varepsilon_{fu} = C_E \varepsilon_{fu}^* \quad (14)$$

where ε_{fu}^* is the maximum guaranteed tensile strain, defined as the average tensile strain (ε_u) less than three times its standard deviation (σ).

The coefficients of environmental reduction provided in Table 3 for GFRP bars are equal to 0.8 and 0.7 for concrete elements non-exposed and exposed to the ground or moisture, respectively.

To account for the creep effect due to sustained load in the design of FRP-reinforced concrete elements, the

Table 3 Environmental reduction factors C_E (after ACI440.1R-03)

Exposure conditions	Type of fiber	C_E
Concrete not exposed to moisture or not in contact with off the ground	Carbon	1.0
	Glass	0.8
	Aramid	0.9
Concrete exposed to moisture or in contact with the ground	Carbon	0.9
	Glass	0.7
	Aramid	0.8

Table 1 Reduction factors for GFRP bars suggested by various codes and design guides

Code	ACI 440.1R-03	CAN/CSA-S6-00	JSCE 1997
Properties	Average -3σ	Average $-\sigma$	Average -3σ
Reduction due to the environmental degradation	C_E 'environmental reduction coefficient' 0.70–0.80	ϕ FRP 'strength factor' 0.75	$1/\gamma_m$ 'factor taking the material into account' 0.77
Reduction due to the sustained load	NA	0.8–1.0	NA
Combined reduction due to the environment and sustained load	0.70–0.80	0.60–0.75	0.77
Design limit taking creep into account ($0.20 f_{fu}$)	0.14–0.16	0.60–0.75	≤ 0.7

Table 2 Typical properties of GFRP reinforcing bars (after ACI 440.1R-03)

Properties	Material	
	Steel	GFRP reinforcing bars
Density lb/ft ³ (g/cm ³)	493.00 (7.90)	77.8 to 131 (1.25 to 2.10)
Longitudinal, CTE $\times 10^{-6}/F$ ($\times 10^{-6}/C$)	6.5 (11.7)	3.3 to 5.6 (6 to 10)
Transverse, CTE $\times 10^{-6}/F$ ($\times 10^{-6}/C$)	6.5 (11.7)	11.7 to 12.8 (21 to 23)
Nominal yield stress, ksi (MPa)	40 to 75	N/A
Tensile strength, ksi (MPa)	70 to 100 (483 to 690)	70 to 230 (483 to 1600)
Elastic modulus $\times 10^3$ (GPa)	29 (200)	5.1 to 7.4 (35 to 51)
Yield strain, (%)	1.4 to 2.5	N/A
Rupture strain, (%)	6.0 to 12.0	1.2 to 1.3

material properties are further reduced. The combined reduction factors for mechanical and environmental effects on GFRP bars used as reinforcement in concrete vary between 0.14 and 0.16.

The tests carried out by Greenwood^[68] on FRP bars under a combination of axial tension and air, salt water, or concrete pore water conditioning showed residual stress corresponding to 44, 27.1, and 14.8% of their initial guaranteed strength, respectively, for the equivalent of 50 years of service at 23°C. Greenwood also obtained the partial factors related to the creep in a saline or an alkaline environment. For E-glass fiber bars, these values were 0.44, 0.62, and 0.34, respectively. An approximate value of 0.45 was obtained for traditional GFRP bars, which is higher than the value recommended by ACI440.1R-03. A recommended factor that takes into account the negative effect of the alkaline environment is 0.34, whereas the factor recommended by ACI440.1R-03 is 0.7. Therefore, it appears that the partial factors of ACI440.1R-03 do not reflect the data generated in Greenwood's study. On the other hand, the combined reduction factor for E-glass fiber bars is equal to 0.15, which is close to 0.14 recommended by the ACI440.1R-03 guide. It can be concluded that ACI440.1R-03 does not consider the differences in performance between the various GFRP bars available for civil engineering applications.

The majority of researchers consider the current codes and guidelines conservative, and it is therefore necessary to adopt a more rational approach to evaluate the materials. Byars *et al.*^[17] proposed a new approach, which takes into account the parameters likely to have a negative influence on the FRP bars during service life. These factors are related to the environment surrounding the concrete, moisture content, temperature, and the stress level. Byars *et al.* assumed that this approach is less conservative, because it is based on real field data. Thus, a coefficient of 0.5 was suggested to account for the negative environmental effects on the GFRP bars. Moreover, they proposed to limit the sustained stress to 30 and 25% of the initial strength for 50 and 100 years of service, respectively.

5.2 PERFORMANCE OF GFRP-REINFORCED CONCRETE BEAMS

The serviceability of reinforced concrete beams is generally controlled by two principal parameters:

- (1) **Crack width:** the presence of large cracks is not aesthetic and can damage or deteriorate the concrete structure, because of high moisture ingress and the risk of reaction between the alkalis and the aggregate,
- (2) **Deflection:** it must be within the acceptable limits imposed by the use of the structure.

The flexural behavior of GFRP-reinforced concrete elements shows less stiffness and wider cracks than

those of concrete elements reinforced with steel using a similar reinforcement ratio, because of the lower modulus of the GFRP bars compared to steel. Consequently, the design of GFRP-reinforced concrete beams is controlled by the deflection and cracking at the service load.

The flexural design of steel-reinforced concrete beams and slabs normally limits the stress in the bars to 60% of its yield stress under the service loads; this limits the strain in the bars to approximately 1200 microstrain ($\mu\epsilon$). This level of strain is meant to limit the crack width in concrete to 0.3–0.5 mm. Since the GFRP bars have an ultimate stress of between 600 and 800 MPa (1.5–2 times the ultimate stress of steel), their lower elastic modulus (0.2 times that of steel) forces the use of much lower levels of stress in order to maintain the crack width within an acceptable range, i.e. below 0.7 mm. The current level of deformation for GFRP bars under service loading is limited by the ISIS-M03-01 design manual to 2000 microstrain.

A more conservative restriction on the permissible stress in FRP bars is imposed by ACI-440.1R-03 to avoid the failure of the bars under sustained load or creep. For fatigue strength, a maximum of 20% of the design stress, corresponding to only 14% of the guaranteed tensile strength of the bars, is allowed. The Canadian code CAN/SCA-S806-02^[15] recommends a maximum of 30% for the guaranteed tensile strength. The Canadian Highway Bridge Design (CHBDC) states that the maximum stress in GFRP bars under loads at SLS shall not be more than 0.25 (CHBDC Clause 16.8.2). Table 4 summarizes the permissible stresses provided by the various guidelines and codes.

5.3 CRACK WIDTH

The crack width in reinforced concrete beams depends on several parameters, such as thickness of the concrete cover and strain level in the reinforcement. Since FRP reinforcing bars are non-metallic, they are immune to environmental corrosion. Therefore, ACI 440.1R-03 allows a maximum crack width of 0.5 and 0.7 mm for external and indoor exposure, respectively.

When the maximum strain in the FRP reinforcing bars under tension exceeds 1500 microstrain, the CAN/CSA-S806-02 code defines a cracking parameter Z given by the following equation:

$$Z = k_b \frac{E_s}{E_f} f_f^3 \sqrt{d_c A} \quad (15)$$

where k_b is a bond coefficient equal to 1.2 for deformed bars, A the effective surface of the concrete section in mm^2 , d_c the effective depth of the section (larger than 50 mm), E_f the elastic modulus of the FRP reinforcement, and E_s the elastic modulus of steel. f_f is the stress calculated for the FRP reinforcement under service load according to eq. (16). This stress cannot

Table 4 Allowable stress Limits for GFRP bars recommended by North-American guidelines and codes

Reference	Equation	Notes	Admissible stress (MPa)
ACI 440.1R-03	$0.20 f_{fu}$	f_{fu} : design stress including the environmental reduction factor	$78^{*,\dagger}$
CAN/CSA-S806-02 (Cl. 7.1.2.3)	$0.30 f_{fu}$	f_{fu} : tensile failure stress	167^{*}
CAN/CSA-S6-00 (Cl. 16.8.2)	$F_{SLS} \cdot f_{FRPu}$	F_{SLS} : reduction factor = 0.25 for GFRP f_{FRPu} : specified stress	140^{*}
ISIS-M03-01	$\varepsilon = 0.002$	equivalent stress-strain [*]	$84^{\ddagger}$

^{*} For a 12.7 mm GFRP bar with: ultimate tensile stress = 639 MPa ($\sigma = 27$ MPa) and guaranteed (specified) tensile stress = 558 MPa

[†] Design stress includes a 0.7 environmental reduction factor applied to guaranteed tensile

[‡] Assumed modulus of elasticity = 42 GPa

exceed 30% of the ultimate stress of the bars[15]:

$$f_t = \frac{M_{ser}}{A_{fd} \left(1 - \frac{k_b}{3}\right)} \quad (16)$$

where A_f is the cross-sectional area of the FRP bars in tension and M_{ser} is the bending moment at service loads.

The code limits the Z value to 38 000 and 45 000 N/mm for external and indoor exposure, respectively. It is not required to calculate Z if the level of strain in the FRP bars under tension does not exceed 1500 microstrains irrespective of fiber type.

5.4 CREEP AND FATIGUE LIMITS

In order to avoid the creep rupture of the FRP reinforcement under sustained or cyclic loading, the allowable stress level must be adjusted accordingly. Since these stresses are related to the elastic behavior of an FRP-reinforced concrete section, they can be calculated with an elastic analysis, as illustrated in Fig. 9 and according to eq. (17)[13]. In this equation, M_s is a service moment corresponding to all the sustained loads:

$$f_{t,s} = M_s \frac{n_f d (1 - k)}{I_{cr}} \quad (17)$$

where I_{cr} is the moment of inertia of the cracked section, $I_{cr} = [b(kd)^3/3] + n_f A_f (d - kd)^2$, in which A_f is the cross-sectional area of the FRP reinforcement, k the ratio of the effective depth to the elastic depth of the neutral axis, $k = \sqrt{2\rho_f n_f + (\rho_f n_f)^2} - \rho_f n_f$, ρ_f the FRP reinforcement ratio, and n_f the ratio of the elastic modulus of the FRP bar to the elastic modulus of the concrete.

The permissible stress limits in terms of creep and fatigue as recommended by ACI 440.1R-03 guide for FRP reinforcement are indicated in Table 3.

6 Concluding Remarks

A comprehensive review of the literature on durability of FRP bars indicates a significant increase

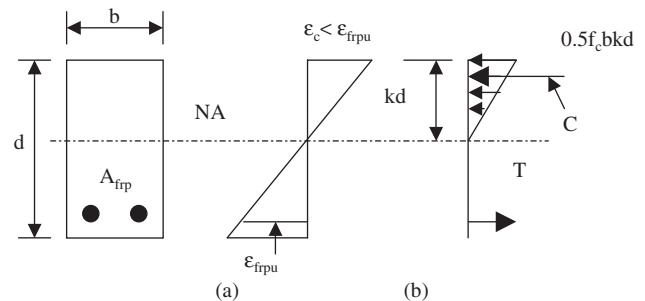


Fig. 9 GFRP-reinforced concrete section: (a) profiles of the distribution of strain; and (b) stress in an elastic state

in the number of studies in this area in the last decade. The majority of the studies focused on GFRP bars for two reasons: the inherent vulnerability of the glass fiber to alkaline environment, which is the situation in concrete structures, and the lower cost of the GFRP bars compared to CFRP bars.

A clear shift in the conclusions of the durability studies was observed. While the early studies done on first-generation GFRP bars condemned the bars as having low durability and highly prone to alkaline corrosion, the durability of the latest generation of GFRP bars is greatly improved. The advancement in the manufacturing process and the application of surface coating barriers on the bars has enhanced their resistance to attack by surrounding mediums.

The review revealed a wide range of techniques used to evaluate the long-term properties of the GFRP bars. Accelerated ageing methods had elevated temperature as a common factor to expedite the degradation mechanisms of the bars; temperatures up to 72°C were used. In addition, several aggressive mediums, such as de-ionized water, alkaline solutions, and salt water, were used in conjunction with, or in isolation from, elevated temperatures. The intention was to simulate the field exposure of the bars, but at a higher rate of reaction. A small number of studies have evaluated the long-term behavior of the bars under sustained stress; this is mainly due to the difficulty of this operation. Recent studies have applied high levels of sustained stress to GFRP, combined with elevated temperature

and aggressive mediums. The residual mechanical properties of the bars after conditioning were slightly affected after a conditioning duration equivalent to 80 years of field exposure.

The relationship between laboratory conditioning methods and life in the field was the focus of several mathematical models. All models used the average ambient temperature at the geographic location of the tests as the datum for the equivalency models. The elevated temperature used in the laboratory was incorporated into the model to transfer the duration of laboratory exposure to real time in the field. However, no correlation was made with the type of medium used or level of stress applied to the bars during conditioning. The absence of test standards and rational models are points in need of immediate attention.

The durability tests conducted by the authors and others on the latest generation of GFRP bars subjected to stresses higher than the design limits, combined with aggressive mediums at elevated temperatures, have concluded that the strength reduction factors adopted by current codes and guidelines are conservative. These factors were based on few test results that were carried out on the early generations of GFRP bars, which have now substantially evolved.

Acknowledgements

The authors acknowledge the financial support from the Natural Science and Engineering Research Council of Canada (NSERC), The Fonds Quebecois pour la Recherche en Nature et Technologie (FQRNT, Quebec), the Canadian Foundation for Innovation (FCI), the Network of Centers of Excellence ISIS-Canada, and the Intelligent Materials and Systems Institute (IMSI) of Université de Sherbrooke.

References

- [1] **Durability of Concrete Structures.** *Canadian Civil Engineer*, May 2004.
- [2] **Intelligent Sensing For Innovative Structures, ISIS-M03-01.** Reinforcing concrete structures with fibre reinforced polymers. The Canadian Network of Centres of Excellence, ISIS Canada, University of Manitoba, Winnipeg, Manitoba, Canada, 2001, 81p.
- [3] **Tannous FE.** Durability of AR non-metallic reinforcement bars and prestressing tendons. *Thèse doctorale*. Département de génie civil et de génie mécanique, Université d'Arizona, 1997, 287p.
- [4] **Bedard C.** Composite reinforcing bars: assessing their use in construction. *Concrete International* 1992; **14**(1): 55–59.
- [5] **Benmokrane B, El-Salakawy EF, Desgagné G & Lackey T.** Building a new generation of concrete bridge decks using FRP bars. *Concrete International* 2004; **26**(8): 84–90.
- [6] **El-Salakawy E, Benmokrane B & Desgagné G.** FRP Composites Bars for the Concrete Deck Slab of Wotton Bridge. *Canadian Journal of Civil Engineering* 2003; **30**(5): 861–870.
- [7] **Huckelbridge A & Eitel AK.** Preliminary performance observations of FRP reinforced concrete bridge deck. Special Publication Field Applications of FRP reinforcement: Case Studies. American Concrete Institute (ACI), Farmington Hills, Michigan, 2003: 121–138.
- [8] **Nanni A & Faza S.** Designing and constructing with FRP Bars: an emerging technology. *Concrete International* 2002; **24**(11): 53–58.
- [9] **Stone D, Nanni A & Myers J.** Field and Laboratory Performance of FRP Bridge Decks, *Proceedings*, Porto, Portugal, FRP Composites in Construction, CCC 2001. In Figueras J, Juvandes L, and Furia R (eds), 2001: 706–710.
- [10] **Bradberry TE.** Concrete bridge decks reinforced with fiber reinforced polymer bars. *Transportation Research Record 1770*, Transportation Research Board, National Research Council, Washington DC, 2001: 94–104.
- [11] **Gangarao HVS, Thippeswamy HK, Kumar SV & Franco JM.** Design, construction and monitoring of the first FRP reinforced concrete bridge deck in the United States. *Proceedings of the 3rd International Symposium on Non-Metallic (FRP) Reinforcement for Concrete Structures, (FRPRCS-3)*, Sapporo, Japan, vol.1, 1997: 647–656.
- [12] **Harries KA, Porter ML & Busel JP.** FRP materials and concrete—research needs. *Concrete International* 2003; **25**(10): 69–74.
- [13] **American Concrete Institute, ACI 440.1R-03.** *Guide for the Design and Construction of Concrete Reinforced with FRP Bars*. American Concrete Institute, Farmington Hills, Michigan, 2003, 41p.
- [14] **Canadian Standards Association, CAN/CSA-S6-00.** *Canadian Highway Bridge Design Code*. CSA International, Rexdale, Ontario, Canada, 2000, 734p.
- [15] **Canadian Standards Association, CAN/CSA-S806-02.** *Design and Construction of Building Components with Fibre Reinforced Polymers*. Canadian Standards Association, Rexdale, Ontario, 2002, 177p.
- [16] **Japanese Society of Civil Engineers.** *Recommendations for Design and Construction of Concrete Structures using Continuous Fiber Reinforced Materials*. JSCCE, Tokyo, Japan (on CD)1997.
- [17] **Byars EA, Waldron P, Dejke V & Demis S.** Durability of FRP in concrete—current specifications and a new approach. In Teng J-G (ed.), *FRP Composites in Civil Engineering*, vol. 2. Elsevier, Amsterdam, 2001: 1497–1507.
- [18] **Karbhari V, Zhao L, Murphy K & Kabalnova L.** Environmental durability of glass reinforced composites-short term effects. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Canada, 1998: 513–524.
- [19] **Katsuki F & Uomoto T.** Prediction of deterioration of FRP rods due to alkali attack. In Taewer L (ed.), *Proceedings of the 2nd International Conference on Non-Metallic Reinforcement for Concrete Structures, (FRPRCS-2)*, Ghent, Belgium, E&FN SPON, London, 1995: 82–89.
- [20] **Benmokrane B & El-Salakawy E (eds).** Durability of fiber reinforced polymer (FRP) composite for construction (CDCC'02). *Proceedings of the 2nd International Conference*, Montreal, Quebec, Canada, 2002, 715p.
- [21] **Porter ML & Barnes BA.** Accelerated durability of FRP reinforcement for concrete structures. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Canada, 1998: 191–202.
- [22] **Sen R, Mullins G & Salem T.** Durability of E-glass/vinylester reinforcement in alkaline solution. *ACI Structural Journal* 2002; **99**(3): 369–375.
- [23] **Nkurunziza G, Benmokrane B, Debaiky AS & Masmoudi R.** Effect of creep and environment on long-term tensile properties of glass FRP reinforcing bars. *4th International Conference on Advanced Composite Materials in Bridges and Structures*, Calgary, Alberta, 2004 (on CD-Rom).
- [24] **Gaona FA.** Characterization of design parameters for composite reinforced concrete systems. *Ph.D. Thesis*. Texas A&M University, 2003, 251p.
- [25] **American Concrete Institute, ACI 440K.** *Guide Test Methods for Fibre Reinforced Polymers (FRP) for Reinforcing or Strengthening Concrete Structures*. American Concrete Institute, Farmington Hills, Michigan (in Committee).
- [26] **Helbling CS & Karbhari VM.** Environmental durability of E-glass/vinylester composites under the combined effect of moisture. *Temperature and Stress, Proceedings of the 2nd International Conference on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC'02)*, Montreal, Québec, Canada, 29–30 May 2002: 247–258.
- [27] **Benmokrane B, Wang P, Ton-That T, Rahman H & Robert J.** Durability of glass fibre reinforced polymer reinforcing bars in concrete environment. *ASCE Journal of Composite for Construction* 2002; **6**(2): 143–153.
- [28] **Alsayed S & Alhozaimy A.** Effect of high temperature and alkaline solutions on the durability of GFRP bars. *1st International Conference on Durability of Fibre Reinforced Polymer Composites for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998.

- [29] **Chong KP.** Durability of composite materials and structures. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998: 1–12.
- [30] **Gerritse A.** Assessment of long term performance of FRP bars in concrete structures. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Canada, 1998: 285–298.
- [31] **Vijay PV & Gangarao HVS.** Accelerated natural weathering of glass fibre reinforced plastic bars. *Proceedings of 4th International Conference on Fibre-Reinforced Plastics for Reinforced Concrete Structures FRPRCS-4*, Baltimore, USA, 1–4 November 1999: 605–614.
- [32] **Karbhari VM.** Durability of FRP composites for civil infrastructure myth, mystery or reality. *Advances in Structural Engineering* 2003; **6**(3): 243–256.
- [33] **Litherland KL, Oakley DR & Proctor BA.** The use of accelerated ageing procedures to predict the long term strength of GRC composites. *Cement and Concrete Research* 1981; **11**: 455–466.
- [34] **Liao K, Schutheisz C, Brinson C & Milkovich S.** Environmental durability of fiber reinforced composites for infrastructure applications: a brief overview. *Proceedings of the Fourth ITI Bridge NDE Users Group Conference*, National Institute of Standard and Technology, Absecon, 1995, 6p.
- [35] **Chin JW, Nguyen T & Aoudi K.** Effects of environmental exposure on fiber reinforced plastic (FRP) materials used in construction. *Journal of Composites Technology and Research* 1997; **19**(4): 205–213.
- [36] **Malvar LJ.** Durability of composites in reinforced concrete. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998: 361–372.
- [37] **Tannous FE & Saadatmanesh H.** Durability of AR glass fiber reinforced plastic bars. *Journal of Composites for Construction* 1999; **3**(1): 12–19.
- [38] **Micelli F, Nanni A & Tegola A.** Effects of conditioning environment on GFRP Bars. *22nd SAMPLE Europe International Conference*, CNIT Paris, 2001, 13p.
- [39] **Chu W & Karbhari VM.** Characterization and moisture and alkali effects on E-glass/vinylester composites. *Proceedings of 2nd International Conference on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC'02)*, Montreal, Québec, Canada, 2002: 359–369.
- [40] **Phillips MG.** Prediction of long-term stress-rupture life for glass fibre-reinforced polyester composites in air and in aqueous environments. *Composites* 1983; **14**(3): 270–275.
- [41] **Benmokrane B, Rahman H, Ton-That M-T & Robert J-F.** Improvement of the durability of FRP reinforced for concrete structures. *1st International Conference on Durability of Fiber Reinforced Polymer For Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998: 571–584.
- [42] **Devalapura RK, Greenwood ME, Gauchel JV & Humphrey H.** Evaluation of GFRP performance using accelerated test methods. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998: 107–116.
- [43] **Swit G.** Durability of stressed E-glass fibre in alkaline medium. In *Recent Developments in Durability Analysis of Composite Systems*, Reifsnider and Verchery (eds) Cardon, Fukuda, Balkema, Rotterdam, 2000: 473–476.
- [44] **Khennane A & Melchers RE.** Mathematical modelling of environmental stress corrosion for glass fibre reinforced polymers. *Research Report No.203.03.2001*, University of Newcastle, Australia, 2001.
- [45] **Khennane A & Melchers RE.** Durability of glass polymer composites subject to stress corrosion. *ASCE Journal of Composite for Construction* 2003; **7**(2): 109–117.
- [46] **Nkurunziza G, Masmoudi R & Benmokrane B.** Effect of sustained tensile stress and temperature on residual strength of GFRP composites. *Proceedings of 2nd International Conference on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC'02)*, Montreal, Quebec, Canada, 2002: 347–358.
- [47] **Mufti A, Onofrei M, Benmokrane B, Banthia N, Boufiza M, Newhook J, Bakht B, Tadros G & Brett P.** Studies of concrete reinforced with GFRP specimens from field demonstration structures. In Hamelin P et al. (eds), *3rd International Conference on Composites in Construction*, Lyon, France, 11–13 July 2005, accepted for publication. (on CD)
- [48] **Canadian Standards Association, Addendum to Section 16, CAN/CSA-S6-00.** *Canadian Highway Bridge Design Code*, CSA International, Rexdale, Ontario, Canada (R2005), 706p.
- [49] **American Concrete Institute.** *State-of-the-art-report on FRP Reinforcements for Concrete Structures*, ACI 440R-96, Detroit, Michigan, 1996.
- [50] **Uomoto T.** Development of new GFRP with high alkali resistivity. *4th International Conference on Advanced Composite Materials in Bridges and Structures ACMBS-IV*, Calgary, Alberta, 2004 (on CD-Rom).
- [51] **Jungkuist DA.** Simulation of enviro-mechanical durability for life prediction of E-glass/vinyl ester composites using a bridge service environment. M.Sc. Thesis, Virginia Polytechnic Institute and State University, 2000.
- [52] **Vijay PV, Ganga Rao HVS & Kalluri R.** Hygrothermal response of GFRP bars under different conditioning schemes. *1st International Conference on Durability of Fiber Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1998: 243–252.
- [53] **Neville AM.** *Properties of Concrete*, 4th edn. John Wiley, New York, 1996, 844p.
- [54] **Valter T & Ralejs T.** Durability and service life prediction of GFRP for concrete reinforcement. In Burgoyne C (ed.), *Proceedings of 4th International Conference on Fibre-Reinforced Plastics for reinforced Concrete Structures FRPRCS-5*, University of Cambridge, Vol. 1, 2001: 505–514.
- [55] **Yilmaz VT & Glasser FP.** Reaction of alkali-resistant glass fibres with cement. Part I: Review, assessment, and microscopy. *Glass Technology* 1991; **32**(3): 91–98.
- [56] **Tannous FE & Saadatmanesh H.** Environmental effects on the mechanical properties of E-glass FRP rebars. *ACI Materials Journal* 1998; **95**(2): 87–100.
- [57] **Tannous FE & Saadatmanesh H.** Durability of AR glass fibre reinforced plastic bars. *Journal of Composites for Construction* 1999; **3**(1): 12–19.
- [58] **Mercier JP & Maréchal E.** *Chimie des polymères, Synthèse, réactions, dégradations*, Presses Polytechniques et Universitaires Romandes, Université Pierre et Marie Curie, Paris VI, 1993, 448p.
- [59] **Sonawala SP & Spontak RJ.** *Degradation kinetics of glass-reinforced polymers in chemical environments*. Department of Materials Science and Engineering, North Carolina State University, Raleigh, NC 27695-7907, USA, 1996: 4745–4756.
- [60] **Gerritse A.** Durability criteria for non-metallic tendons in an alkaline environment. In Neale KW & Labossière P (eds.), *Proceedings of 1st International Conference on Advanced Composite Materials in Bridges and Structures, ACMBS-I*, Sherbrooke, PQ, Canada, 1992: 129–137.
- [61] **Theodore JR.** *Engineered materials Handbook, Engineering Plastics*. vol. 2. Ohio: ASM International, 1998: 272–275.
- [62] **Dejke V & Tepfers R.** Durability and service life prediction of GFRP for concrete reinforcement. In Burgoyne C (ed.), *Proceedings of 4th International Conference on Fibre-Reinforced Plastics for Reinforced Concrete Structures FRPRCS-5*, vol. 1, University of Cambridge, Cambridge, 2001: 515–520.
- [63] **Benmokrane B, Ton-That MT, Robert JF & Mukhopadhyaya P.** Durability studies of composite GFRP C-bar reinforcing rods. *Technical Progress Report No.8* submitted to Marshall Industries Composites, inc., Department of Civil Engineering, Faculty of Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, 1999: 105p.
- [64] **Thomason JL.** The interface region in glass fibre-reinforced epoxy resin composites: 3. characterization of fibre surface coatings and the interphase. *Composites* 1995; **26**(7): 487–498.
- [65] **Mallick PK.** Fiber-reinforced composites, materials, manufacturing, and design. Ph.D. Thesis, Department of Mechanical Engineering, University of Michigan-Dearborn, Dearborn, Michigan, 1988, 469p.
- [66] **Chin JW, Haight MR, Hughes WL & Nguyen T.** Environmental effects on composites matrix resins used in construction. *1st International Conference on Durability of Fibre Reinforced Polymer for Construction (CDCC'98)*, Sherbrooke, Quebec, Canada, 1988: 229–241.
- [67] **Phillips MG.** Prediction of long-term stress-rupture life for glass fibre-reinforced polyester composites in air and in aqueous environments. *Composites* 1983; **14**(3): 270–275.
- [68] **Greenwood M.** Creep-rupture testing to predict long-term performance, durability of fiber reinforced polymer (FRP) composites for construction. *2nd International Conference Durability of Fibre Reinforced Polymer for Construction (CDCC'02)*, Montreal, Québec, Canada, 29–30 May 2002: 203–212.
- [69] **Devalapura RK, Gauchel JV, Greenwood ME, Hankin A & Humphrey T.** Long-term durability of glass-fiber reinforced polymer composites in alkaline environments. *Proceedings of 3rd International Conference on Fibre-Reinforced Plastics for Reinforced Concrete Structures FRPRCS-4*, vol. 2, Baltimore, MD, 1997: 83–90.
- [70] **Sirois C.** *Étude sur la durabilité des armatures en fibre de verre sous tension dans un milieu alcalin*, Mémoire de maîtrise, Université de Sherbrooke, Département de génie civil, Sherbrooke, Canada, 1998, 114p.
- [71] **Robert JF.** Évaluation de la durabilité en milieu alcalin des armatures à base de fibres de verre pour les structures de génie civil en béton armé,

Mémoire de maîtrise, Université de Sherbrooke, Département de génie civil, 1999, 148p.

[72] **Laoubi A.** Étude de la durabilité de poutres en béton armé de barres en matériaux composites de PRF. *Thèse doctorale*, Sherbrooke, Département de génie civil, Université de Sherbrooke, 2003, 205p.

[73] **Nkurunziza G.** Performance of Glass FRP Bars as Reinforcement for Concrete Structures under the Effect of Sustained Loads and Elevated Temperature

in Humid and Alkaline environment. *Ph.D. Thesis*, Department of Civil Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, 2004, 190p.

[74] **Wang P.** Effect of moisture, temperature, and alkaline on durability of E-glass/vinyl Ester reinforcing bars. *Ph.D. Thesis*, Department of Civil Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, 2005, 154p.

[75] **Pultrall Inc.** *ISOROD Composite Reinforcing Rods*. Technical Data Sheet, Pultrall Inc. Thetford Mines, PQ, Canada, 2000, 7p.

Gilbert Nkurunziza

Structural design engineer
Teknika Inc
Sherbrooke, Quebec, Canada

Ahmed Debaiky

NSERC Industrial Fellow
A. S. Composite
48-3608 St-Charles
Kirkland, Quebec, Canada H9H C3C

Patrice Cousin

Post-doc Fellow
Department of Civil Engineering,
Faculty of Engineering,
University of Sherbrooke,
Sherbrooke, Quebec, Canada J1K 2R1

Brahim Benmokrane

NSERC Research Chair,
Department of Civil Engineering,
Faculty of Engineering,
University of Sherbrooke,
Sherbrooke, Quebec, Canada J1K 2R1
E-mail: Brahim. Benmokrane@usherbrooke.ca